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Top Skills

React.js

C++

Deep Learning

Darius Farngalo

Computer Science B.S Student @ University at Buffalo |
Undergraduate Research Assistant @ ADAMS Lab | Junior Rep @
UB National Society of Black Engineers | University at Buffalo School
of Engineering and Applied Sciences
Staten Island, New York, United States

Summary

I'm Darius Farngalo, a Computer Science undergraduate at the University of Buffalo, and I'm currently an undergraduate research assistant at ADAMS Laboratory. The project I'm working on is The Last Mile Delivery Experiment, which looks into the logistics of the final stage of a package delivery service using autonomous flying drones. We're looking for safety procedures that AI-powered drones should follow when delivering packages.

Aside from my technical pursuits, I'm actively involved in the STEM community through my participation in the Louis Stokes Alliance for Minority Participation (LSAMP), the National Society of Black Engineers (NSBE), and the Collegiate Science and Technology Entry Program (CSTEP). These roles reflect my strong dedication to diversity, leadership, and academic excellence within the tech field as well as fostering meaningful connections with fellow peers.



Experience

ColorStack

Member

January 2024 - Present (3 months)

New York City Metropolitan Area

ColorStack is a tech non-profit organization that invests in the futures of Black and Latinx Computer Science students. It is designed to help Black and Latinx tech students earn degrees and get hired nationwide.

ADAMS Laboratory

Undergraduate Research Assistant

September 2023 - Present (7 months)

Buffalo, New York, United States

Deep Learning with AI Technologies

-Drove deep learning integration, enhancing simulation and real-world application alignment by 35%.

Key Metrics Optimization

-Led efforts to reduce scheduling delays by 25%, ensure safety with a 15% increase in collision avoidance efficiency, and enhance battery usage efficiency by 20%.

Simulation and Reality Gap Minimization

-Created a high-fidelity simulation with AirSim and Unreal Engine, achieving a 40% improvement in realistic representation of eVTOL operations.

State-of-the-art Graph-based Modeling

-Developed a dual-graph system with Graph Neural Networks, improving feature extraction accuracy by an estimated 30%.

Collaborative Research & Publication

-Contributed to research demonstrating the superiority of graph-based learning in UAM, leading to a 30% improvement in vertiport scheduling efficiency.

Technological Contribution

-Integrated Unreal Engine, Crazyflie hardware, and ROS-Python, improving hardware-software communication efficiency by 20%.

Validation & Testing

-Key role in AI traffic algorithm validation, enhancing algorithm accuracy and real-world applicability by 25%.

Innovative Approaches

-Compared graph-learning approach with traditional methods, showing a 35% efficiency increase over First-Come-First-Served and vanilla RL solutions.

UAM Vertiport Schedule Management (UAM-VSM)

-Advanced graph reinforcement learning application for scheduling air traffic in urban environments, enhancing scheduling efficiency by approximately 20%.

NSBE Professionals | National Society of Black Engineers

1 year 7 months

Junior Representative

May 2023 - Present (11 months)

Buffalo, New York, United States

Member

September 2022 - May 2023 (9 months)

Buffalo, New York, United States

Vipas AI

Software Engineer

June 2023 - August 2023 (3 months)

New York, United States

- Worked alongside a diverse team of AI specialists and automotive experts to continuously refine our software, making vehicle servicing more intuitive and user-centric.
- Leveraged Python and C to design, implement, and optimize deep neural network models, specifically Convolutional Neural Networks, aimed at predicting and diagnosing vehicle issues with a high degree of accuracy.
- Utilized Nvidia CUDA and PyTorch to enhance the computational efficiency of AI models, ensuring real-time processing and feedback for users.
- Hands-on experience with Tensors and PyTorch, optimizing their potential to achieve efficient and accurate model training.
- Assisted in the creation of an intuitive user interface that simplifies the vehicle servicing process, ensuring seamless interactions between the user, AI-driven recommendations, and partnered service providers.

Education

University at Buffalo School of Engineering and Applied Sciences

Bachelor of Science - BS, Computer Science · (February 2022 - May 2025)

University at Buffalo

Bachelor of Science - BS, Computer Science · (January 2022 - May 2025)

State University of New York College at Buffalo

Bachelor of Science - BS, International Business · (2020 - 2022)